BROAD INSTITUTE PRISM MULTIPLEXED CELL LINE SCREENING

PRISM MTS Submission Information

PRISM MTS screens are standard offerings for DMSO-soluble compounds in a single-agent format. These screens are run using our standard design: 5-day assay, 900+ cell lines, test agents run at 8 pt. dose for single-agents, with 3-fold dilutions in triplicate. Collaborators select the top screening dose as a part of the screen submission process. Additional workflow information can be found on the PRISM website.

Test Agent Requirements

- We require at least 150µL at 1000X the top screening dose in 100% DMSO.
  - i.e. If your top screening dose is 10µM, you will submit a 10mM stock.
- The collaborator is responsible for ensuring that the test agent is soluble in 100% DMSO at the stock concentration and for QC'ing test agents ahead of submission.
  - We do not have the bandwidth to do any optimization for, or re-running of, test agents that do not perform well in PRISM. Any test agents submitted will be run 'at-risk'
  - We are unable to solubilize powdered stock of compounds for collaborators.

Compound Handling Workflow

NOTE *Lab work for PRISM screens (from compound submission to lysate generation) can take up to 2 months and compounds may go through multiple freeze/thaw cycles. It is important that compounds submitted for consortium screens be tested for stability and QCed prior to submission.*

1. Test agents are shipped or delivered to the Broad Institute prior to the submission deadline. They are stored according to the conditions specified on the collaborator submission form until step 2 begins (-20°C, 4°C or room temperature (RT)).
2. After the submission window has closed, compounds must be registered in the Broad's Compound Management LIMS. If compounds were frozen prior to this point, they will be thawed and stored at RT during the registration process (this can take up to 24 hours). Compounds will be transferred to the proper automation-friendly vials and submitted in batches to the Compound Management team, who will give each compound a BRD# and store the compounds in a large -20°C freezer until assay-ready plates (ARPs) are made.
3. Compounds will then be thawed in order to generate ARPs. The compounds are added to source plates and diluted in 100% DMSO using a Hamilton liquid handler (several hours).
4. Source dilution plates are then transferred acoustically to destination ARPs using an Echo liquid handler (~18hr.). The final DMSO concentration is 0.1% (1000X dilution from the 100% DMSO source plate).
5. Once all ARPs are dispensed, plates are vacuum-sealed and stored at -20°C until the day of screening.
6. Assay-ready plates (ARPs) are thawed before cell plating day, labeled with pool information, and scanned into an ELN for tracking and metadata creation.
7. During plating, cells are dispensed into the ARPs using a Combi liquid dispenser with the following conditions:
   a. Adherent Lines- RPMI-10% FBS 1250 cells/well
   b. Mixed Lines- RPMI-20% FBS, 2000 cells/well
   c. Suspension Lines- RPMI-20% FBS, 3000 cells/well
8. Cell-containing ARPs are incubated for 5-days before being lysed at the endpoint.

Updated: June 2026